A Practical Guide to Linear regression

TOPIC -- LINEAR REGRESSION

$$\hat{y} = X\beta + \varepsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

-- 01 --

arg max β → l (D , β →) = arg max β → (n log ■ 1 2 π σ − 1 2 σ 2 ∑ i = 1 n (y i − β → · x i →) 2) = arg min β → ∑ i = 1 n (y i − β → · x i →) 2 = arg min β → L (D , β →) = β ^ → {\displaystyle {\begin{aligned}{\underset {\vec {\beta }}{\mbox{arg max}}}\,l(D,{\vec {\beta }})&={\underset {\vec {\beta }}{\mbox{arg max}}}\left(n\log {\frac {1}{{\sqrt {2\pi }}\sigma }}-{\frac {1}{2\sigma ^{2}}}\sum _{i=1}^{n}\left(y_{i}-{\vec {\beta }}\,\cdot \,{\vec {x_{i}}}\right)^{2}\right)\\&={\underset {\vec {\beta }}{\mbox{arg min}}}\sum _{i=1}^{n}\left(y_{i}-{\vec {\beta }}\,\cdot \,{\vec {x_{i}}}\right)^{2}\\&={\underset {\vec {\beta }}{\mbox{arg min}}}\,L(D,{\vec {\beta }})\\&={\vec {\hat {\beta }}}\end{aligned}}}

-- 02 --

ε {\displaystyle {\boldsymbol {\varepsilon }}} is a vector of values ε i {\displaystyle \varepsilon _{i}} . This part of the model is called the error term, disturbance term, or sometimes noise (in contrast with the "signal" provided by the rest of the model). This variable captures all other factors which influence the dependent variable y other than the regressors x. The relationship between the error term and the regressors, for example their correlation, is a crucial consideration in formulating a linear regression model, as it will determine the appropriate estimation method. Fitting a linear model to a given data set usually requires estimating the regression coefficients β {\displaystyle {\boldsymbol {\beta }}} such that the error term ε = y − X β {\displaystyle {\boldsymbol {\varepsilon }}=\mathbf {y} -\mathbf {X} {\boldsymbol {\beta }}} is minimized. For example, it is common to use the sum of squared errors ■ ε ■ 2 2 {\displaystyle \|{\boldsymbol {\varepsilon }}\|_{2}^{2}} as a measure of ε {\displaystyle {\boldsymbol {\varepsilon }}} for minimization.

-- 03 --

Y = l n β 1 X + ■ {\displaystyle Y=ln\beta _{1}X+\epsilon } In the above equations, the coefficients (parameters) are non-linear. A model with exactly one explanatory variable is a simple linear regression; a model with two or more explanatory variables is a multiple linear regression. This term is distinct from multivariate linear regression, which predicts multiple correlated dependent variables rather than a single dependent variable. In linear regression, the relationships are modeled using linear predictor functions whose unknown model parameters are estimated from the data. Most commonly, the conditional mean of the response given the values of the explanatory variables (or predictors) is assumed to be an affine function of those values; less commonly, the conditional median or some other quantile is used. Like all forms of regression analysis, linear regression focuses on the conditional probability distribution of the response given the values of the predictors, rather than on the joint probability distribution of all of these variables, which is the domain of multivariate analysis. Linear regression is also a type of machine learning algorithm, more specifically a supervised algorithm, that learns from the labelled datasets and maps the data points to the most optimized linear functions that can be used for prediction on new datasets. Linear regression was the first type of regression analysis to be studied rigorously, and to be used extensively in practical applications. This is because models which depend linearly on their unknown parameters are easier to fit than models which are non-linearly related to their parameters and because the statistical properties of the resulting estimators are easier to determine. Linear regression has many practical uses. Most applications fall into one of the following two broad categories:

-- 04 --

Adaptive Estimation If we assume that error terms are independent of the regressors, ε i ⊥ x i {\displaystyle \varepsilon _{i}\perp \mathbf {x} _{i}} , then the optimal estimator is the 2-step MLE, where the first step is used to non-parametrically estimate the distribution of the error term.

-- 05 --

History Isaac Newton is credited with inventing "a certain technique known today as linear regression analysis" in his work on equinoxes in 1700, and wrote down the first of the two normal equations of the ordinary least squares method. The Least squares linear regression, as a means of finding a good rough linear fit to a set of points was performed by Legendre (1805) and Gauss (1809) for the prediction of planetary movement. Quetelet was responsible for making the procedure well-known and for using it extensively in the social sciences.

-- 06 --

Y i = β 0 + β 1 X i 1 + β 2 X i 2 + … + β p X i p + ■ i {\displaystyle Y_{i}=\beta _{0}+\beta _{1}X_{i1}+\beta _{2}X_{i2}+\ldots +\beta _{p}X_{ip}+\epsilon _{i}}

-- 07 --

X {\displaystyle \mathbf {X} } may be seen as a matrix of row-vectors x i · {\displaystyle \mathbf {x} _{i\cdot }} or of n-dimensional column-vectors x · j {\displaystyle \mathbf {x} _{\cdot j}} , which are known as regressors, exogenous variables, explanatory variables, covariates, input variables, predictor variables, or independent variables (not to be confused with the concept of independent random variables). The matrix X {\displaystyle \mathbf {X} } is sometimes called the design matrix. Usually a constant is included as one of the regressors. In particular, x i 0 = 1 {\displaystyle x_{i0}=1} for i = 1 , … , n {\displaystyle i=1,\ldots ,n} . The corresponding element of β is called the intercept. Many statistical inference procedures for linear models require an intercept to be present, so it is often included even if theoretical considerations suggest that its value should be zero. Sometimes one of the regressors can be a non-linear function of another regressor or of the data values, as in polynomial regression and segmented regression. The model remains linear as long as it is linear in the parameter vector β. The values xij may be viewed as either observed values of random variables Xj or as fixed values chosen prior to observing the dependent variable. Both interpretations may be appropriate in different cases, and they generally lead to the same estimation procedures; however different approaches to asymptotic analysis are used in these two situations.

-- 08 --

Example Consider a situation where a small ball is being tossed up in the air and then we measure its heights of ascent hi at various moments in time ti. Physics tells us that, ignoring the drag, the relationship can be modeled as